

NVIDIA CORP NVDA

April 04, 2023 - 2:39am EST

by differentiatedfractal31415

			2023	2024
Price:	279.65	EPS	0	0
Shares Out. (in M):	2,507	P/E	0	0
Market Cap (in \$M):	701,100	P/FCF	0	0
Net Debt (in \$M):	-1,260	EBIT	0	0
TEV (in \$M):	699,840	TEV/EBIT	0	0

Description

Preamble: NVDA was posted earlier as a short here by glgb913. I am writing from the other side, to present my long thesis on NVDA. The stock has rallied hard since our underwriting and pitching it at this time is seemingly difficult, especially given the YTD run-up and outperformance relative to the rest of the market. We are long-term focused patient investors, and I can't find a better idea to write up before the submission deadline here, illustrating my variant perception of the business. I editorialize most of the technical concepts to help advance the debate on the business fundamentals. **(Read the disclaimer and risks sections towards the end of this post as you conduct your own due diligence. This post is not a recommendation to buy or sell a security but rather intended for a discussion of key bull and bear debates, based on research of public information, including company websites and regulatory filings. This post does not constitute professional or investment advice. Past performance is not indicative of future results. Consult with a licensed professional if needed before making any investment decisions. Investing in public markets can be subject to severe short-term and long-term risks, including resulting in permanent loss of entire invested capital. This post represents my personal opinions and not that of any other entity. I and/or entities that I advise can change views and positioning, at any point of time for any reason, or lack thereof, without any notice. I and/or entities that I advise are not and will not be held liable for any outcomes, including any indirect or consequential damages. I can be 100% wrong. This article has no guarantee to be accurate, complete, or current, and cannot be relied on as such. Please conduct your own due diligence.)**

I believe in a concentrated yet balanced portfolio, including some best-of-breed names along with some structural hedges with other ideas, targeting a risk-adjusted positive skew. Nvidia was underwritten as a key ingredient for our growth exposure over the next 5 years in this new compute cycle. I'd argue it is much less of a cyclical semi and has better structural moats than several software names. Given a product-cycle-driven and potentially catalyst-rich backdrop in a new compute regime, I go over my investment thesis, providing some context around the underlying tech. I address a set of key bear arguments and hope this is helpful to think through the key debates.

Investment Thesis

(i) Nvidia GPUs are general-purpose accelerators that should continue to take share in datacenter.

First, here's the context. Per an empirical law proposed by Gordon Moore (RIP), transistor count on semiconductor chips doubled every two years until recently. Another empirical law, Dennard Scaling was based on power density being constant when transistor size decreases, so clock frequencies scaled higher, and chips got faster. But that stalled after 2005 because of heat dissipated at small dimensions.

Then came the concept of multiple cores (chip components that execute instructions). Central Processing Units (CPUs) use multi-cores (128 cores in some latest ones in the market) and target single-thread performance. Graphics Processing Units (GPUs) use many smaller cores (some NVDA GPUs have 10,000+ now) and were built initially for gaming applications. But the main difference is that GPUs are designed to maximize throughput with single-instruction multiple-thread (SIMT) architecture, which makes parallel processing possible [1]. So now you can simultaneously render pixels on the screen to show you that cat, everything everywhere all at once, without having to wait in a serial fashion i.e. one pixel at a time.

Adopting a fabless model with foundry partners such as Taiwan Semiconductor (TSMC), Nvidia developed GPUs demonstrating high throughput, measured as 8-bit tera operations per second (TOPS), doubling every year from Kepler (2012) at 4 to H100 (2022) at ~4000, more than that predicted by Moore's law (doubling every 2 years).

A 'big bang' for the field of AI happened in 2012, when a grad student team from U. Toronto won the ImageNet, a major computer vision contest, with a model trained on NVDA GTX 580 GPUs given their programmability and high floating-point performance on the Fermi architecture. (One of the researchers later became the cofounder of OpenAI.) NVDA then developed more general-purpose (GP) GPUs and improved chip architectures over the last decade. Tensor cores were introduced in Nvidia's Volta (2017) to exploit sparsity, a feature of matrix multiplications involved in training large language models. Tensor cores offer lower precision (good enough outside of supercomputing), and hence higher throughput. Nvidia also developed transformer engines, which include high-level programming with linear algebra libraries such as TensorRT, for machine learning models, including LLMs such as GPT.

The period 1960-2010 saw a lot of ups and downs for AI, including the notable AI winter of the 1970-80s. The doubling period for compute measured in the number of operations (in units of petaflop/s x days) was every 2 years. After 2012, models got bigger and NVDA GPUs facilitated accelerating the compute doubling time to every 3.4 months [1]. Starting with use cases such as recommender systems and social network feeds, NVDA scaled to 30%+ market share (of revenues, much less in units) in datacenter processor revenues within just a decade of the early GPGPUs.

NVDA has gained about 430 bps of datacenter revenue share/year in the last 5 years. NVDA's dominant position in the accelerated compute market is likely to continue. Hyperscaler spending on accelerators is estimated to still be a low portion (<10%) of their capex and should inflect upwards in this new cycle. We expect NVDA to continue taking share from CPUs over this compute cycle. We model NVDA's datacenter revenues to grow faster than the current consensus CAGR estimate for the next 5 years.

(ii) Nvidia is not just a chip maker; its unique co-design approach integrating hardware and software has resulted in a heterogenous-compute system ideal to ride the current inflection in AI cycle.

While Nvidia's chips garner a lot of attention (given consumer gaming was a major use case previously), what's more important is their systems-level solution with a full-stack approach to hardware and software for accelerated compute. Over the last decade, given its highly profitable gaming segment, NVDA invested in R&D at an average 24% of revenues which funded growth in GPGPUs. NVDA developed a series of GPU architectures that evolved over the years, advancing not only compute (throughput) but also storage (memory capacity and memory bandwidth). Think of memory capacity as the number of books your backpack can carry and memory bandwidth as how much you can open it so the books can be transferred fast to the locker. NVDA GPU memory bandwidth increased 8x in a span of 10 years, with typical CPUs still at an order of magnitude below GPUs. The ability to perform parallel processing and the memory advancements helped GPUs become the computing workhorse on which LLMs grew exponentially.

In an industry ever-focused on Moore's law, which clearly slowed down in the last decade due to limitations of physics in manufacturing at small scales, Nvidia also focused on Amdahl's law, which is about solving weakest links or bottlenecks in the system level solution. NVDA developed the NVLink fabric for fast interconnect between system components, such as CPUs and GPUs. A highly relevant acquisition from 2020 in this context was Mellanox, which provides InfiniBand high-bandwidth interconnects from system to system, and a host of other networking (ethernet switching) solutions.

Early on, NVDA abstracted the middle layer of software (the low-level language interacting with the hardware) into a unified architecture layer (across all generations of GPUs) called CUDA. Software CUDA platform provides developers end-to-end workflow solutions and comes with high switching costs. *AI/ML systems are fundamentally stochastic and not deterministic*, whether from initializing weights and biases in a neural network or attention mechanisms in a transformer. This new software regime was dubbed as Software 2.0 by Andrej Karpathy (former Tesla Sr. Director that returned to OpenAI recently) at the AI Summit 2018. In contrast to Software 1.0, coding is <10% of machine learning systems; data workflow management is the pain point [2]. If you run the same code twice you may get different solutions, and managing data workflow becomes critical for debugging as well as explainability.

As AI gets operationalized across enterprises, it becomes critically important for developers to have end-to-end solutions for managing workflows. The switching costs in this new regime are much higher than typical SaaS businesses, as more than performance or cost savings, what matters more is developer comfort and inertia. Over the years NVDA developed custom accelerated libraries, frameworks, SDKs, and a comprehensive compilation stack on the CUDA GPU computing platform. With a robust ecosystem of 4M developers, NVDA's CUDA helps form a structural competitive advantage that is hard to replicate. CUDA remains a strong driver for GPU performance and ASP uplift at each new GPU architecture.

(iii) While NVDA's likely dominance in training is understood, NVDA's value proposition for inference workloads is underappreciated.

Inference is critical as it is performed real-time (i.e low latency at the prompt or the customer will search elsewhere, for example) in production unlike training — which is performed as batch processing that can last several days. Per Amazon at their AWS:Reinvent 2022 "for every \$1 spent on training, up to \$9 is spent on inference." So, as inference costs per query reduce over time and adoption rises, it is likely inference market can grow to be bigger than its current state. At GTC 2023, NVDA provided 4 optimized solutions for inference, L4, L40, H100 NVL, and Grace Hopper. NVDA's CUDA/Triton inference engines offer critical capabilities such as model/hardware optimization, and quantization.

Currently, industry estimates are at about 60-70% of current inference workloads being performed on CPUs. From our research, we believe that as LLM and other multi-domain use cases gain traction, the market is going to shift away from CPUs as they cannot meet GPUs in price performance for handling the compute requirements of LLMs.

A more important consideration is sustainability, with news headlines around power utility infrastructure not being able to keep pace with data center growth, for example, in one of the major data center corridors [3]. As datacenters optimize their power envelope, NVDA's performance/watt advantage over peers should stand out, and any energy savings would likely be reinvested for further acceleration of workloads in the datacenter.

(iv) OpenAI is talked about as the AWS-like critical infrastructure software play of this AI cycle; but I think the market is underestimating the potential for NVDA as an Operating System layer for the AI stack.

NVDA recently laid out AI foundation models as a service, including pre-trained large models of Nemo, Bio-Nemo, Picasso. It also developed the DGX Cloud to be deployed leveraging the infrastructure of cloud service providers (MSFT, GOOG, ORCL). In GTC 2022, NVDA talked at length about AI Operating System helping enterprises with adoption of AI/M, with CUDA as the low-level programming layer, pre-trained models and end-to-end data management solution including (i) integration with enterprise tensorflow/pytorch workloads, (ii) base/fleet command for edge/distributed workloads and (iii) domain-specific toolkits. With new AI foundation models and a cloud-agnostic AI cloud service DGX Cloud for custom GPU instances, NVDA's full-stack solution facilitates an operating system layer for enterprises, with potential for monetization over est. 50M enterprise servers worldwide.

(v) Rebound in gaming and strength in auto segment: NVDA's gaming revenues are guided to rebound to \$2.5B/quarter run rate potentially later this year. In other segments, at GTC 2023, NVDA updated the auto 6-year design pipeline to \$14B (vs \$11B/\$8B in previous two GTCs). NVDA-powered automotive computing systems for fleets of Jaguar Land Rover (build from 2025) and Mercedes (partnership announced in 4QFY23 call), should support incremental growth longer term.

My rebuttals to some key bear arguments:

(i) Bear argument: "Training demand is mostly pulled forward and there will be fewer large models, likely lowering compute required."

Per OpenAI's paper released with GPT-4 model (model size undisclosed) was already trained in Aug 2022. If the trends of Google's Chinchilla and Meta's LLaMA model papers were to continue, models are likely getting mostly smaller, with larger models maintained by only a few larger players. While we agree with that, larger models are not going away for sure; they tend to be more data efficient (fewer tokens needed to train) and are needed to derive smaller models. If the industry gets to using more quantities of smaller models, we believe that they may have to be trained on large corpora of data tokens to reduce model log-losses. Such tokens could include proprietary datasets, for e.g. Bloomberg recently released a research paper documenting the development of BloombergGPT. Nvidia's DGX cloud is now available on hyperscaler clouds for enterprises to train their own models. The rise of domain-specific enterprise

use-cases could continue supporting compute demand tailwinds.

(ii) Bear argument: "The bigger growth segment of AI market is likely inference, which is a competitive market, and NVDA's edge in training is missing here."

As discussed in the investment thesis points, inference is potentially a land grab for accelerators. NVDA is likely to gain a substantial share in this growing segment given (i) low latency, better performance per watt at scale, and resulting TCO advantage of its GPUs (ii) incumbency of training loads already present on NVDA GPUs, and (iii) various software and hardware optimizations available with the Nvidia CUDA/Triton inference engine.

(iii) Bear argument: "Datacenter capex is likely pressured given overbuild during Covid."

While we acknowledge datacenter capex digestion is likely given the capacity build-up in the covid years, we think this dynamic is well understood from the last earnings at most datacenter plays. Datacenter capex dedicated to accelerated server spend is likely to inflect upwards and we believe capex estimates are likely to be revised up over the next 5-10 years of this compute cycle, as structural demand shifts may become clearer to hyperscalers.

(iv) Bear argument: "Hyperscalers will continue developing their own custom silicon solutions and NVDA's growth and margins will be pressured."

At GTC 2023, Jensen announced Google Cloud, Microsoft Azure, and Oracle OCI as hyperscaler partners for deploying DGX Cloud. **Google Cloud was announced as a partner for deploying NVDA L4 inference chips for LLMs.** This positive surprise should assuage concerns around Google TPUs taking away the shine from NVDA GPUs. While we wait to see how this debate eventually plays out, for those bears that argue hyperscalers could get competitive in chip-making and inhibit the rise of DGX cloud, our counter would be:

(i) GPUs are likely to perform better than custom silicon for a wider range of general-purpose workloads.

(ii) Hyperscalers would benefit from being "utilities" [4] drawing diverse workloads to the platform, selling their commoditized compute and storage to, for example, new crop of AI-native startups. For example, Snowflake, as a cloud-agnostic data warehouse that leverages hyperscale cloud infrastructure, grew while competing against Amazon Redshift, Microsoft Azure Synapse Analytics, and Google BigQuery.

(v) Bear Argument: "Newer software frameworks such as OpenAI Triton, Pytorch 2.0, or TensorFlow are a threat."

I think it is unlikely existing workloads can be transferred easily given the end-to-end solution for debugging and explainability (similar to version control of Software 1.0) that's critical in AI/ML workflows. With the several accelerator libraries added each year, e.g. CuDNN, CuLitho, CuOpt, and CuQuantum, CUDA's value proposition has been increasing as an all-stop shop for AI/ML developers. CUDA is free for users and aimed to increase the adoption of NVDA hardware.

From our industry checks, the feedback we got was that (i) it seems hard to usurp Nvidia's 15+ year advantage of providing the bells and whistles of AI/ML training and inference workloads, unless the new software competitors are proven to be much better and hardware is much cheaper. (ii) CUDA offers critical low-level programming without which it is hard, if not impossible, to get NVDA GPUs run optimally. Ultimately the debate can be settled perhaps by which hardware system-level scale-out solution has the best performance (by price and per watt).

(vi) Bear Argument: "AI hype cycle is no different than that we have seen for metaverse or crypto."

While I agree there's a lot of excitement for AI, it is hard to brush it off as a hyped-up craze when we consider the origins of this compute cycle and developments since 2018 which make automating creative output. ChatGPT Plus queries running slow at peak times in the day and faster at night reminds us of the dial-up modem days of internet of the 1990s. Per an AI Infrastructure survey [5] by run.ai, 89% of companies face compute allocation issues regularly. A recent article [6] from The Information mentioned that even Microsoft employees seem to be having a hard time getting access to GPUs.

Unlike metaverse or crypto, the AI use-cases are tangible and with massive reach, including a knowledge worker base of 1.5 B, and large and small enterprises adopting AI. While some normalization of demand is reasonable to expect over the next 5-10 years in this compute cycle, we also expect to see the rise of several new use-cases: for example, LLMs and edge inference could breathe a fresh life into iPhone cycle (re-vamp of Siri, more custom filters likely with multi-modal GPT-4).

Valuation

Nvidia's datacenter revenues exceeded 50% of its total revenues in CY2022. Over 2017-2022, datacenter revenues grew at a 51% CAGR. Street models we see seem to grow this at 22% CAGR over the next 5 years. We model a base case CAGR of 30% for this estimate. We believe the higher CAGR is reasonable to expect when you consider the drivers within inference, as well as training. Hyperscaler capex grew at ~28% CAGR over the last 5 years. The trajectory of this capex going forward may have near-term uncertainty given overbuild during covid. But AI spend is likely a priority as evidenced from recent AI initiatives at major hyperscalers. We model hyperscaler capex as growing at 15% CAGR over 2022-27E, a relatively lower pace than that during 2017-22. Our modeled NVDA datacenter revenue comes to within mid-teens percentage of our hyperscaler capex forecast.

The split between training and inference revenues for Nvidia is undisclosed. For the next 5 years, the debate is on what inference market could be relative to that of training. NVDA CEO estimated at GTC 2023 that inference is likely to grow to be the same size as training, which informs our model assumption for inference. We model 70%/30% mix of CPU/accelerated for inference workloads now eventually shifting to a 30%/70% given our conviction from our research on the need for GPUs in inference, with some share loss factored to account for accelerators potentially gaining traction. For training, we are with street in assuming GPU workload accelerating at 25% CAGR, with other accelerators modeled as growing at a faster pace from their smaller base, and non-parallelizable CPU workloads growing at a slower pace.

My base case EPS 5-yr CAGR comes to be over 30% under this framework, reaching earnings power in the range of \$13.5-\$15. Assuming an exit NTM P/E of 35x, we underwrite about 15% IRR from current levels. With discounting at a rate of 10% for 2 years, we get to \$390 2024E PT. If NVDA's software solutions get monetized over the next five years to even low single-digit percentage of AI enterprise TAM (50M servers at \$3K fee/year) presented in GTC 2022, there

could be an additional upside. I discuss potential catalysts and risks to my thesis below.

Risks to my investment thesis, where I can be wrong:

(i) If rates stay higher for longer under potentially persistent high inflation, growth names could be under pressure and NVDA's valuation multiple could contract. In the longer term, stock returns follow earnings, and our variant perception is on earnings potential, not a market multiple. We think continued execution in datacenter/auto and normalization of gaming could start an upwards earnings revision cycle for NVDA, which can potentially attract incremental buyers in any severe drawdown. From our research, it seems competitive risks are likely limited given NVDA's hardware/software and ecosystem advantage. While we are not macro investors, portfolio hedge ideas based on growth or several other factors could be critical for portfolio construction and risk management.

(ii) As of 4QF23, Nvidia reported inventory at a high 212 days, with management guiding for normalization in the second half of this year. We will be closely following earnings to track this metric.

(iii) Gaming may take longer to normalize and pose a growth/margin drag.

(iv) AI/ML markets may be competitive and NVDA may cede share to other accelerators (Cerebras, Sambanova, Graphcore, Intel Habana) and peer GPU makers (AMD and Intel), and FPGAs (Altera and Xilinx). Our base case factors a modest GPU share loss to these peers, but to the best of our knowledge, Nvidia benefits from its CUDA advantage, robust ecosystem, and fast pace of innovation (e.g. arriving at new use-case-specific products such as L4, L40, or H100 NVL) at scale.

(v) Nvidia had >20% of sales exposure to China as of last year. Nvidia developed products such as A800 and H800 to satisfy export control laws. Any export control restriction headlines may adversely affect sentiment.

References:

[1] OpenAI, "Two distinct eras of compute usage in training AI systems," May 2018

[2] Sculley et al, "Hidden Technical Debt in Machine Learning Systems," NeurIPS 2018

[3] R. Miller, "Dominion Resumes New Connections, But Loudoun Faces Lengthy Power Constraints," Data Center Frontier, Sep 2022

[4] B. Moore, "We will begin to think of the cloud providers as "utility" rather than "solution" providers," from "Computing and data infrastructure in '22," Kleiner Perkins, Jan 2022.

[5] 2023 [State of AI Infrastructure Survey](#), run.ai Accessed March 2023

[6] A. Holmes and K. McLaughlin, "Microsoft rations access to AI hardware for internal teams," The Information, March 15, 2023

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Catalyst

Potential catalysts to watch for:

- Datacenter :
 - Nvidia navigating through supply chains for meeting H100/A100 demand
 - Adoption of L4/L40/H100 NVL for inference, and of Grace (CPU), Bluefield (DPU), Grace Hopper (CPU+GPU), DGX cloud
 - More disclosure around revenue model/partnership with CSPs for DGX cloud, foundation models
 - ASP uplift in the aggregate portfolio as higher ASP products come to the market
 - Rebound in demand from China hyperscalers for products A800, H800
 - Pickup in enterprise demand for Nvidia GPUs / Software / AI Operating system
- Other Segments
 - Rebound of gaming demand as supply chains get back to normal
 - Continued strength in auto / omniverse demand

Messages

Subject	Software question
Entry	04/04/2023 03:30 PM
Member	Bismarck

Hey thanks for the writeup.

We have done a lot of work on AMD specifically and are not the smartest on the GPU software side. Understand the basics and why AMD is lagging there specifically. What I would like to better understand, though, is why you think NVDA's advantage thru CUDA will be relevant to the bulk of industry players by purchase quantity. What I mean by this is - sure, your random AI startup is going to 100% use CUDA given the libraries, support, etc. For the hyperscalers, I think you can ask the inverse, why WOULD they remain sole sourced on NVDA (and CUDA) and not port their software over to open source libraries to enable them to use other hardware. It seems like a no brainer to me. It would definitely be worth their time, given the discounts they could get on AMD hardware (which performs just as well from a hardware perspective - if you look at AMD's latest accelerator).

Maybe I do not understand the intricacies so well, but porting over large models from CUDA is being made easier with OpenAI Triton. Obviously small devs don't care / don't have the resources to even bother, but companies purchasng billions of hardware w/ teams of devs certainly would? And they are actively sponsoring these open software alternatives - which is a direct sign of their intent, no?

Subject	Re: Software question
Entry	04/04/2023 09:31 PM
Member	yxd0950

<https://www.reuters.com/technology/meta-launches-ai-software-tools-help-speed-up-work-blog-2022-10-03/>

There's a view from the AI community that open-sourced software will get better to avoid capture. No one was trying hard to come up with an alternative to CUDA but things could have changed given how important AI has become.

What Meta is doing is by using AI interference to port CUDA code into alternatives. You'd think it's not a hard training problem to solve since there's plenty CUDA code to train on.

Subject	China risk
Entry	04/05/2023 12:49 PM
Member	DO EM GO

Differentiatedfractal31415,

I think you're write is very helpful for investors like me who haven't done a lot of technical work in semiconductors and is a reminder that the analysis of this industry is outside my core competency.

My feedback in this This response will be less about the technical merits of NVDA vs. AMD and others but more on the geopolitical risk it faces. The multiple here is astoundingly high in my view given the China risk I feel is a sizable tail risk. This pretty much goes for AAPL, TSLA etc that have very large Chinese exposure. My thoughts are as follows.

- 1) The US and China are clearly engaged in a Cold war of sorts. China's ascendancy is being confronted by the US head on, especially militarily. The US has moved to restrict the most powerful chips China needs to achieve military parity with the US and the wherewithal to build by forcing ASML others to stop selling to China.
- 2) China has thus far tried to get around the export restrictions but the noose is getting tighter.
- 3) China has allied itself with global thugocracies like Russia, Iran, Saudi Arabia etc. It seems increasingly clear the world is falling into a Cold War 2.0 split.
- 4) China knows it needs certain Western tech (mainly high end Chips) to continue to try develop militarily to insulate its Communist/autocratic police state from US liberal democracy push. This is the most important tech they need that is becoming harder and harder to obtain.
- 5) Meanwhile US consumer brands sold in China at high margins does nothing in terms of helping the Chinese goal of US military parity. In fact, allowing the Apples and Tesla's of the world and NVDA to a certain extent to sell to Chinese consumers at huge gross margins only enriches the US at the expense of local Chinese producers such as Huawei and BYD Auto. Those gross margin dollars could be used by local Chinese champions to boost R&D and try to build markets in the Middle East, Russia (thugocracies around the world).
- 6) China clearly needs export markets and the argument against shutting out Western companies in an abrupt manner seems low as it would suffer economically. However, it can inflict tremendous pain on the Western World as we are clearly dependent on those cheap exports to keep prices down. Furthermore, Chinese control of lithium supply and more importantly rare Earth metals could become a weapon to use against the in an increasingly bitter cold war. I hope I am wrong but is this not where we seem to be headed?
- 7) The bogus investigation into Micron is likely the first of many small steps toward an eventual closing of Chinese markets to US multinationals. I have no idea how quickly this happens but at some point one would think the Chinese will reach a point of belief that a complete withdrawal from the Western World is advantageous to them. Xi seems to believe their system is better and we will collapse before they do.

Conclusion: Am I an idiot and is all of this completely wrong? Clearly the performance of NVDA, AAPL etc indicates the market doesn't view this as a real risk. Of course I realize that security prices are not set by rational actors seeking to protect their personal capital so perhaps the current multiples are not telling us much of anything...

GLTA

Subject	Nice
Entry	05/24/2023 07:29 PM
Member	ajp579

Obviously the big story-growth names aren't well-loved on here so congrats on what I thought was a thoughtful writeup (gave it a much higher than avg quality score but admittedly low performance score) as well as the strong performance so far.

You were pretty sober in your writeup but here's an attempt to more brashly summarize the bull case from here:

1. Literally every tech company is overhauling their stack with AI right now – look at MS CoPilot and Adobe firefly which were announced / released (respectively) just THIS WEEK. Between the arms race to build ever-bigger foundation models and inference just beginning to take off in earnest, demand for GPU compute is probably the biggest growth story in the entire economy right now
2. Designing GPUs is really hard – With the rest of Big Tech focused on the application layer (rightly so), the CUDA architecture should give NVDA at least a couple years of being the 'only game in town' for building scaled AI applications
 1. Some commentary below about 'can't ppl just use AI to port CUDA to other formats' – feels like low-level machine code is still too tough for this to be practical
3. This is a great 50% EBITDA margin business that only needs to grow ~200% from here to grow into a 20-handle ~market multiple – certainly not 'cheap' by any traditional metric but seems fairly achievable

Thoughts on this?

Subject	Re: Nice
Entry	05/24/2023 08:31 PM
Member	Shoe

So now \$11 in EPS looks pretty doable, run-rating \$8 annualized EPS with 'substantial ramp up' coming in 2H. Dunno how sustainable that is, but it seems pretty sustainable for a few quarters at least, which gives you some runway.

NVDA is the purest AI play right now, at 35x EPS, it's not unreasonable here for what clearly is an integral part of the next generational technological advancement

Long onlies will have to chase to get equal weight or overweight. Others will chase it as well.

My guess is it goes higher.

Maybe others will buy the other tangential AI plays like ANET, AVGO, MRVL, TSM. MRVL is an interesting laggard play and has been hampered by inventory issues which will resolve soon.

Plus META, GOOG, AMZN, ORCL, MSFT can keep running. BTW, ADBE's new generative AI tech looks pretty slick

Subject	Re: Nice
Entry	05/25/2023 09:02 AM
Member	ahnuld

not to nit pick but I dont think there will continue to be a race to "build ever-bigger foundational models". Its unclear right now if adding more parameters gets better performance than fine tuning existing models on specific domain knowledge.

This was published yesterday, 65B parameters, 99% chatbot ability of GPT4

<https://huggingface.co/blog/4bit-transformers-bitsandbytes>

I think there may be bigger demand from companies wanting more compute on prem (in this case 64GB ram is enough) to fine tune and not want to feed everything to azure/aws.

Subject	Update
Entry	05/25/2023 06:11 PM
Member	differentiatedfractal31415

Nvidia long thesis seems to be playing out as I had outlined in my thesis, albeit at a pace faster than I expected. Consensus was at ~\$11 for C2027E EPS before the earnings call, and seems like some analysts are now modeling that for C2024E – 3 years earlier than expected. My 2027E EPS moves from \$13.5-15 to \$18-20, and using the framework presented in my thesis, I get to a 2024E PT of \$550+

Jensen didn't quite answer Stacy's inference question on the earnings call yesterday. My thesis is on growth of workloads and share gain in that segment (away from CPUs). The multimodal inference use case (highlighted in the post) was cited on the call, so also the continued tailwinds for training. ServiceNow's domain-specific (ITSM) LLM built on Nvidia's Nemo foundation model and deployed on Nvidia's DGX cloud is an example illustrating my thesis that NVDA can be a compelling play across multiple layers of the AI tech stack. Mellanox remains a crown jewel of an acquisition that is helping NVDA gain share in datacenter.

Thanks for the questions here.

- DO EM GO. China headlines are a risk, agreed, but I wonder if it is prudent to limit the use of A800/H800s at an AI inflection.
- Bismarck, I am eager to follow what AMD has to say at the June event on MI300 as well as ROCm – as a big fan of Lisa Su and the consistent share gains in datacenter that AMD had at the expense of Intel. I model some share loss to AMD and other accelerators – so not quite concerned on that front.
- As for the rise of custom silicon at hyperscalers, recent announcements at Google I/O as well as Meta's adoption of more GPUs are aligned with my thesis that NVDA is likely a must-have for external workloads given the 4M+ developer ecosystem reliant on NVDA's full-stack solution. While Chris Miller has a great title (and great book!), it is not really a Chip War, it is a Systems War when it comes to accelerated computing in the datacenter. One should keep that in mind when considering the prospects for MTIA. It took NVDA ~15 years to be a software powerhouse with low level programming with CUDA and a whole host of high level programming libraries, APIs, compilers, and integrations with several frameworks.
- Yxd0950, thanks for the Reuters article. My research indicates that CUDA kernel programming is a hard and rare skill. Agree with Apj59 that it seems like CUDA could be difficult to be disrupted in an AI interference hackathon.
- Apj59, on valuation, it's hard to pin oneself down to a P/E where the NTM E is underestimated. What works for me is to have a differentiated thesis on earnings potential in outer years and debate all the bear thesis points. Valuation by itself does not make a bear thesis, unless there is an unforced error- and it's hard to be a perma-bear given NVDA's cadence of product innovation. The upside is my model arises from inference being a greater opportunity than the market would like to recognize at this time, and that's one of the reasons for my underwriting. We are at the early stages of a new product cycle in a new tech regime, with estimate revisions like today's likely to come along the way; NVDA remains a core position for me despite said valuation concerns.
- Ahnuld, agree that smaller models could be on rise but the number of parameters is only one degree of freedom when designing transformers and their attention mechanisms. You also have others such as number of data tokens (more to reduce model log-losses and hallucinations), number of epochs (repeats on entire data set), and context window in your LLM query UX. Computational intensity scales exponentially with context window. My thesis on enterprises is that a full stack solution is important for data management and explainability of the AI models. This could perhaps limit liability in decisions made by stochastic models. The last I checked there are fewer than 100 chatgpt plugins on the plus account, and the more this ecosystem develops and the more the likes of ServiceNow's ITSM-like domain-specific models are built from foundation models, training demand will likely have further tailwinds. I have modeled a share loss from the current 90%+ level to other accelerators in training, and will monitor how this share shift evolves as inference inflects upwards and shifts away from CPUs.
- Shoe, thanks for your take, and the good call on Marvel!

On a personal note, I am thankful for having a voice on this platform, and appreciate you listening whether or not you agree with my viewpoints.

Please see the disclaimer included at the end of the original post and risks associated with this name.

Subject	Open AI is Reportedly Heavily GPU-Constrained
Entry	06/02/2023 10:18 AM
Member	ajp579

Found this a very interesting article:

https://website-754fwhahs-humanloopml.vercel.app/blog/open_ai_talk

Looks like OpenAI is hitting compute constraints in BOTH training and inference:

- The number of params in the new 'cutting edge' model is growing more slowly due to compute costs, despite the fact that scaling laws appear to still apply
- The attention mechanism used at inference – the key innovation that made ChatGPT work -- is $O(n^2)$, meaning a 10x longer context window (prompt size) is 100x more computationally expensive. I suppose it's TBD whether a 1million token (word) context window is enough for most commercial applications, but I'd have to imagine there are at least a few use cases that require an LLM to 'reason' over a very large input sequence

Subject Forbes article on stability ai
Entry 06/05/2023 06:58 AM
Member pcm983
<https://www.forbes.com/sites/kenrickcai/2023/06/04/stable-diffusion-emaad-mostaque-stability-ai-exaggeration/>

Subject Re: Forbes article on stability ai
Entry 06/05/2023 10:14 AM
Member jgalt
 why is this relevant?

Subject Re: Re: Forbes article on stability ai
Entry 06/05/2023 11:19 AM
Member ajp579
 Pretty weak hit job if you ask me, here's the rebuttal:

<https://emad.posthaven.com/on-setting-the-record-straight>

If anything, seems bullish for NVDA that the world is so eager to cast AI into the crypto/theranos/wework bucket even on the flimsiest of accusations

Subject Re: Re: Re: Forbes article on stability ai
Entry 06/05/2023 03:38 PM
Member jso1123

This is a great piece of investigative journalism (wish we had more of this versus what journalism due to economic pressure has become). We know underlying any big shift as monumental as the emergence of gen AI are both reality and hype. And the opportunity to get rich attracts fraud. I think we will see many more grifters like this.

Think of the emergence of ecom in the late 1990s - tons of fraudsters/b.s. and hype way ahead of reality, but also in there were a handful of companies that would be massively successful and career-making investments.

This cycle will play out the same way.

Subject Re: Re: Re: Re: Forbes article on stability ai
Entry 06/06/2023 12:57 PM
Member ajp579

Generally agree that there's both reality and hype underlying gen AI. However let me tell you about an even more egregious grifter:

- Avid psychedelic drug user. Refused to cut his hair or wear shoes
- Dropped out of Reed College (nontarget much?) after just one semester (compare that to actually attending Oxford but just failing to pick up your degree)
- Refused to recognize a daughter he had with a woman he was not married to
- All he really did was commercialize a product that his partner actually built

Oh wait – that's Steve Jobs.

We haven't had a truly transformative new technology for awhile so maybe it's worth reminding ourselves that the people who end up commercializing those technologies usually don't fit a traditional mold.

Frankly, ML was a pretty dull field even within academia for decades before the last few years, so I'd guess that anyone with more than a couple YOE in the field probably doesn't have the best commercial instincts.

My suspicion is that this piece has more to do with Emad's commitment to open sourcing most of the AI tech. Gee, who might have financial incentive to go after the guy doing that?

Subject CUDA advantage no more
Entry 06/30/2023 10:38 PM
Member yxd0950

<https://www.semianalysis.com/p/amd-ai-software-solved-mi300x-pricing>

"7 months ago, we described how Nvidia's dominant moat in software for machine learning was weakening rapidly due to Meta's PyTorch 2.0 and OpenAI's Triton. We have also discussed the work MosaicML's has been working on since as far back as last year. With the latest PyTorch 2.0, MosaicML Composer, and Foundry releases, AMD hardware is now just as easy to use as Nvidia hardware.

To date, this was mostly for Nvidia hardware. MosaicML's stack can achieve over 70% hardware FLOPS utilization (HFU) and 53.3% model FLOPS utilization (MFU) on Nvidia's A100 GPUs in large language models **without requiring writing custom CUDA kernels**. Note that Google's stack for PaLM model on TPUv4 only achieved 57.8% HFU and 46.2% MFU. Likewise, Nvidia's own Megatron-LM stack only achieved 52.8% HFU and 51.4% MFU on a 175B parameter model. Mosaic's stack, much of which is open source, is an obvious choice unless every last drop needs to be squeezed out with many dedicated scaling engineers for clusters of 10,000s of GPUs.

Now, MosaicML is going to be able to offer the same with AMD hardware. They have only just gotten their Instinct MI250 GPUs this quarter, but they are already close to matching Nvidia.

We profiled training throughput of MPT models from 1B to 13B parameters and found that the per-GPU-throughput of MI250 was within 80% of the A100-40GB and within 73% of the A100-80GB.

Abhi Venigalla, MosaicML

There is no code change required.

mapping every floating point operation, every GPU command, and every distributed operation like ``torch.matmul()`, `torch.cuda.current_device()`, `inputs.to('cuda:0')`, `torch.dist.all_gather()`, etc.` call to the appropriate ROCm and RCCL operations on the AMD system. See Figure 1 for screenshots of what this looks like. We agree, **it's funny to run `torch.cuda` on an AMD machine, but it works!**

Abhi Venigalla, MosaicML

Subject	Update
Entry	08/23/2023 11:55 PM
Member	differentiatedfractal31415

There's a lot to like in this 2QF24 earnings call. The systems (not just chips) approach of NVIDIA seems to be working, given structural demand tailwinds: 'platform shift' of data centers from general purpose CPU-based compute to accelerated compute, and gen AI applications in this new tech cycle.

Some highlights from the earnings report/call:

- NVDA posted 101% y/y revenue growth this quarter, along with strong cash flow generation: 2QF24 FCF margin 45% (39% if SBC is not added back).
- Days Sales of Inventory: 97 (normalized from the highs of ~225 from 4QF23).
- Data center demand was broad-based with CSPs >50%, followed by consumer internet customers and enterprises/HPC.
- L40S GPUs are repurposed RTX 40 graphics cards, unaffected by CoWoS supply constraints (they don't have HBM). Jensen seems to suggest this is for scaleout of GPUs for a different set of (perhaps lighter) workloads, while H100s will likely be the workhorse for LLMs at the hyperscalers (given their HBM but likely their supply is dictated by CoWoS capacity ramp-up and the supply chain cycle times)
- GH200 Grace Hopper superchip integrated with HBM3e is expected to be available in 2QCY24E
- Partnerships on AI/ML models (Hugging Face), Data warehouses (Snowflake), Enterprise Software platforms (ServiceNow, VMWare), marketing (WPP), Content Creation (Adobe, Getty Images, Shutterstock) help illustrate the strength of the ecosystem and diverse set of applications in focus.

Some modeling considerations:

- Think gross margin estimates in sellside models should go up given H100s being a larger part of the revenue mix (data center at 76% of revenues in 2QF24 and ASPs for H100 likely ~2.7x that of A100s) and their gross margins estimated to be 80%+.
- Operating leverage should continue to help: incremental non-GAAP margins were 75%+ in the quarter and likely 70%+ implied from the guidance given for 3QF24. You perhaps don't need to hire as many sales reps and R&D engineers in the midst of a product mix-driven ~2.7x ASP uplift.

I think there could be a line of sight for \$25 earnings power potentially by CY25E, contingent on maybe \$14-15B+ quarterly run rate of DC revenues. If you apply a 30x NTM P/E you get to a 2024E PT of \$750. I can be wrong if demand levels from CSPs (and consumer internet customers) miss expectations, but that's perhaps unlikely as we seem to be still early in a data center infrastructure spending cycle. Also one needs to believe CoWoS packaging bottlenecks would resolve (Colette mentioned on the call that they expect supply to increase every quarter for CY23E and into CY24E).

Subject	Re: Update
Entry	08/24/2023 12:28 PM
Member	Jyle

Could you explain the logic NVDA is using to justify accelerated computing replacing all CPUs? Feels difficult to believe given most software developed today is built on the basis of CPU instruction sets...shifting 100% to accelerators would require massive overhaul of software today...what's the incremental benefit to having everything on accelerated compute? I think I'm missing the point, but would be interested in hearing an intelligent answer.

Subject	Re: Re: Update
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Entry 08/25/2023 04:14 PM
Member differentiatedfractal31415

You won't get rid of the quarterback. CPU runs the show- that's how GPUs are usually designed to operate. You can't make a brick (serial tasks) fly (parallelize on multiple cores). Platform shift means that you need running backs on jet engines, so you accelerate whatever can be parallelized and get the productivity boost illustrated by Amdahl's law. If you are trying to understand the TAM, a way to think is % of capex of hyperscalers/enterprises allocated to upgrading old/building new datacenters. Hope that helps.

Subject Re: Re: Re: Update
Entry 08/25/2023 09:24 PM
Member IdeaLogue

I get that no one really knows, but how are you thinking about the level of sustainable demand here? Thanks!

Subject Re: Re: Re: Re: Update
Entry 09/04/2023 08:16 PM
Member lightspot

why isn't NVDA the new Cisco? what's different really? Back in 2000s, the capacity was overbuilt because the demand just wasn't yet ready. Broadband speed wasn't fast enough and pricing wasn't cheap enough. Smartphones weren't broadly available, which was a key driver to bring internet access to everyone, anywhere.

So the key question is, why isn't the market doing the same thing now and overestimating demand for AI? What makes anyone think this time is different? ChatGPT which seems to be by far the most advance LLM, isn't yet validated by Enterprise use cases. It is still in "experimental phase". I asked a bunch of tech people I know, and none of the engineers think is a serious tool yet.

Subject Re: Re: Re: Re: Re: Update
Entry 09/04/2023 11:17 PM
Member yxd0950

how is Co-Pilot not a serious tool?

Subject Re: Re: Re: Re: Re: Re: Update
Entry 09/05/2023 10:37 AM
Member lightspot

Copilot has a nice marketing video. But do you know anyone using it extensively? I don't.

Subject Re: Re: Re: Re: Re: Re: Re: Update
Entry 09/05/2023 12:50 PM
Member TheEnterprisingInvestor

how are you guys thinking about some of these arrangements w/ entities like CoreWeave?

Subject Re: Re: Re: Re: Re: Update
Entry 09/05/2023 01:15 PM
Member ajp579

I think key difference between AI and early days of the internet is how much compute is needed for training, which has to be done before any end users can even touch the product (inference).

Also would point out ChatGPT is a product, not an LLM itself (you can switch between multiple LLMs as a user)

In a way lightspot, your comment that end demand isn't there yet is super bullish. If all of this demand so far is just for training, imagine how much more demand there will be once inference actually starts to take off (think the original poster largely made this point). Don't forget about the tens-of-billions of dollar bet Zuck was just making on AR with Meta, which is arguably much further from generating real end user demand. Big tech is basically mandated to plow their billions in monopoly profits into moonshot R&D, and AI suddenly seems like the least farfetched thing they could be working on.

As a very uninformed observer, it seems pretty obvious that eventually NVDA will have a bunch of competition for CUDA and hit a supply glut. But it is far from obvious to me that that day is coming any time soon.

Seems like the next big test for AI and NVDA is MS Office CoPilot coming out later this year – if that's a flop with nontechnical users imagine this hype bubble starts to deflate.

Subject Re: Re: Re: Re: Re: Re: Update
Entry 09/06/2023 11:23 AM
Member lightspot

I agree that the demand for inference could be way way bigger. Here is a question then. Once the models are trained, the market will move to the inference stage. I'd imagine that we will have multiple different applications. Some focused on video, others focused on text, etc. Doesn't make sense the hardware will be customized according to the end user application? I will use the crypto market as an example. Bitcoin was the dominant application (dominant cryptocurrency). Hence, why companies like Bitmain created ASICs and that dominated Bitcoin mining. GPUs were left for the "Alt coins", because no other crypto was big enough for a chip company to specifically design a chip just for that. And most altcoins were all built on top of Ethereum. Isn't the AI inference market end up following similar path? We may have specific hardware customizations for video inference, others for text inference and so on. Effectively making the inference TAM for Nvidia GPU significantly smaller? or diminishing their dominant position as others (Intel/AMD/Arm and the CSPs) being able to create custom hardware for specific inference use cases?

Subject Re: Re: Re: Re: Re: Re: Re: Update
Entry 09/06/2023 07:59 PM
Member Weighing Machine Capital

Well said lightspot. An example of the different optimization profiles for training vs. inference is Tesla's self driving architecture.

Tesla has just implemented a 10,000 Nvidia H100 cluster for its end-to-end neural network V12 FSD beta for training, but, for its inference computer at the edge, in its cars, the inference computer is actually quite simple in its compute capacity/power requirements. Its HW3 chip only runs at 100 watts at 30 frames/second. The reason is it needs to produce millions of these "edge" computers to put in its cars. Tesla needs to keep the cost reasonable (<\$1,000 per inference computer) and the power drain low in order to keep from using too much of the cars' battery for inference.

Inference compute doesn't need the computing power that training compute needs. And importantly, given that there's going to be inference chips in every laptop/mobile phone/etc., there's no way an Nvidia A100 or H100 can go into edge devices that consumers/businesses use for inference. And obviously, the power requirements for GPUs are a deal-breaker for edge devices for inference.

Nvidia will continue to dominate training compute for the next few years, but inference....no way. Additionally, the hyperscalers are all coming up with their own optimized training chips (Google already does 90% of its training on its own TPU architecture; not Nvidia clusters). Nvidia is certainly riding the training compute wave and exercising its first-mover advantage, but as these models shift from training to inference, it will be technically challenging both from a cost perspective and power requirement perspective to use Nvidia GPUs.

Subject Re: Re: Re: Re: Re: Re: Re: Update
Entry 09/06/2023 09:47 PM
Member yxd0950
 Lightspot,

Yes I talked to people and they said productivity doubled because they can work on important business logic more. Copilot is a godsent that dramatically reduces redundant and tedious coding chores programmers hate the most. Think about writing unit tests.

But I agree with you in general that Nvidia is probably overvalued at this point and inference would be a different game. For example, some of the biggest open-sourced LLMs can be run locally for inference on a MacBook, which clearly doesn't use a NVDA card.

Subject Re: Re: Re: Re: Re: Re: Re: Re: Update
Entry 09/07/2023 11:35 PM
Member lightspot

Thank you Weighing Machine for the detailed color. Do you think companies like Intel, AMD and Arm are better positioned for supplying the inference hardware for the edge, since they have more of their chips already going to those end devices vs Nvidia?

For example, Arm has a dominant position in smartphones. Nvidia has 0 GPUs in smartphones. Common sense makes me think that if anyone's going to create a specific inference hardware for smartphones, that it should be Arm. I wonder if you or anyone else see different and why.

Subject Re: Re: Re: Re: Re: Re: Re: Re: Re: Update
Entry 09/08/2023 01:56 PM
Member Weighing Machine Capital

Thanks lightspot,

I think inference chips on edge devices are going to be highly specialized depending on the use case. There's going to be tradeoffs for compute power (flops/sec.) vs. power usage; the more compute power -> the higher the power requirement + cost. There's always tradeoffs in the engineering/design of chips. GPUs consume way too much power to be applicable to edge inference devices and are cost-prohibitive for phones/cars/laptops, robots, etc. AMD/QCOM/Arm are designing specialized chips for inference; especially AMD with their acquisition of Xilinx for specialized programming through its FPGA capabilities. However, these chips will become somewhat commoditized but the key is the software layer that will allow for continuous updates to further optimize for the specific use case.

Subject Re: Re: Re: Re: Re: Re: Re: Re: Re: Re: Update
Entry 09/11/2023 12:24 PM
Member lightspot

Based on what've I read, it seems that AMD MI300 will be very competitive for training against Nvidia grace hopper H100s. My understanding is that memory seems to be the key in order to scale out to thousands of GPUs and parallelize the algorithm in a way that there is very little interdependency between each GPU (thus achieving high utilization rate) when training large models. MI300 has much more memory than H100. People will say that H100 have better performance/efficiencies, but again, based on what I think I'm starting to understand here is that # of F16 or F32 flops doesn't matter as much as memory. Also, AMD may have not released performance metrics yet because its still pre-launch and I guess they normally don't do this before launching new products? So the claim that H100 have higher performance aren't rock solid! Curious if anyone has any feedback on this.

Subject New GPU / AI HW sale restrictions
Entry 10/14/2023 09:35 AM
Member MississippiCo

Scoop: Biden's team to announce new export limits on AI chips to China (axios.com)

Does anybody have a better / more comprehensive source of info on what this may look like? Could obviously have some impact for NVDA (~20% of FY22 revs came from china), but I wonder if anyone has seen any industry or SS work that gives a good overview of current and potential restrictions. GS did a good overview a ~year ago during the initial regs on semicaps but I haven't seen anything since then... Thanks!

Subject Re: New GPU / AI HW sale restrictions
Entry 10/23/2023 01:20 PM
Member MississippiCo

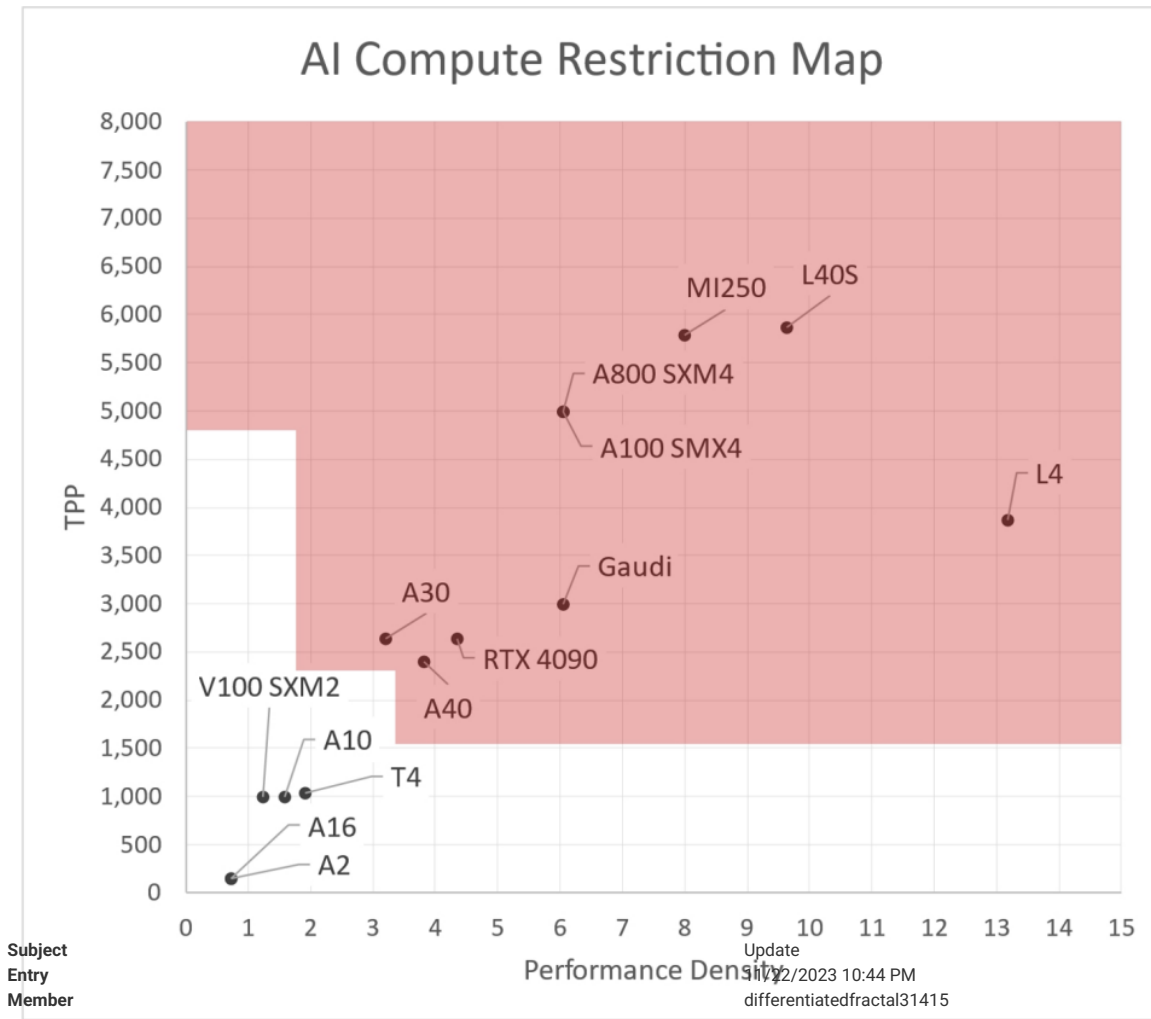
Found a couple good ones from UBS:

Figure 2: AI Compute Restrictions Model

Product		ECCN 3A090 - Results			Process			Datasheet Specifications							TFLOPs (dense)							ECCN 3A090 - Total Breaches			
Manufacturer	Product Name	Requires License	Transistor Count Red Flag	HBM Red Flag	Process Node	Non-planar	CoWoS	Transistors (B)	GPU Name	Logic Die Size (mm2)	Max. Memory (GB)	Memory Type	Memory Bandwidth (GB/s)	NVLink (GB/s)	INT8	FP8	FP16	BFLOAT16	TF32	FP32	FP64	FP64 Tensor Core	Total Breaches	TPP Breaches	PD Breaches
NVDA	A100 SXM4	YES	YES	YES	7nm	YES	YES	54,200	GA100	826	80	HBM2e	2,039	600	624.00	NA	312.00	312.00	156.00	19.50	9.70	19.15	4	4	0
NVDA	L40S	YES	YES	NO	5nm	YES	NO	76,300	AD102	609	48	GDDR6	864	NA	733.00	NA	362.50	362.05	183.00	91.60	NA	NA	5	4	1
NVDA	H100 SXM5	YES	YES	YES	4nm	YES	YES	80,000	GH100	814	80	HBM3	814	900	1979.00	1979.00	989.50	989.50	494.50	67.00	64.00	67.00	7	5	2
NVDA	A800 SXM4	YES	YES	YES	4nm	YES	YES	54,200	GA100	826	80	HBM2e	2,039	400	624.00	NA	312.00	312.00	156.00	19.50	9.70	19.15	4	4	0
NVDA	H800 SXM4	YES	YES	YES	4nm	YES	YES	80,000	GH100	814	80	HBM3	814	400	1979.00	1979.00	989.50	989.50	494.50	67.00	64.00	67.00	7	5	2
NVDA	V100 SXM2	NO	NO	YES	12nm	YES	YES	21,100	GV100	815	32	HBM2	900	300	125.00	NA	31.33	NA	NA	15.67	7.83	NA	0	0	0
NVDA	RTX 4090	YES	YES	NO	5nm	YES	NO	76,300	AD102	609	24	GDDR6x	1,008	NA	NA	NA	82.58	NA	NA	82.58	1.29	NA	1	0	1
NVDA	L4	YES	NO	NO	5nm	YES	NO	35,800	AD104	294	24	GDDR6	300	NA	242.50	242.50	121.00	121.00	30.30	0.49	NA	4	0	4	
NVDA	A2	NO	NO	NO	8nm	YES	NO	8,700	GA107	200	16	GDDR6	200	NA	NA	NA	4.53	NA	4.53	NA	0.07	NA	0	0	0
NVDA	A10	NO	NO	NO	8nm	YES	NO	28,300	GA102	628	24	GDDR6	600	NA	NA	NA	31.24	NA	31.24	NA	0.98	NA	0	0	0
NVDA	A16	NO	NO	NO	8nm	YES	NO	8,700	GA107	200	16	GDDR6	200	NA	NA	NA	4.53	NA	4.53	NA	0.07	NA	0	0	0
NVDA	A30	YES	YES	YES	7nm	YES	YES	54,200	GA100	826	24	HBM2e	933	NA	330.00	NA	165.00	165.00	82.00	10.30	5.20	10.30	4	0	4
NVDA	A40	YES	NO	NO	8nm	YES	NO	28,300	GA102	628	48	GDDR6	696	112.5	299.30	NA	149.70	NA	74.80	37.40	0.55	NA	3	0	3
NVDA	T4	NO	NO	NO	12nm	YES	NO	13,600	TU104	545	16	GDDR6	320	NA	130.00	NA	65.00	NA	NA	8.14	NA	NA	0	0	0
AMD	M1250	NO	YES	YES	6nm	YES	Similar	58,200	CDNA 2.0	724	128	HBM2e	3,277	NA	NA	NA	362.10	NA	45.26	NA	45.26	NA			
AMD	MI300*	YES	YES	YES	5nm	YES	YES	146,000	DNA 3.0	1017	128	HBM3	3,277	NA	NA	1253.50	383.00	NA	47.87	NA	47.87	NA	3	2	1
Intel	Gaudi2	YES	NO	YES	7nm	YES	Similar			826	96	HBM2e	2,450	NA	NA	NA	508.56	NA	254.28	NA	NA	NA	2	2	0
Intel	Gaudi	YES	NO	YES	16nm	YES	Similar			495.6	32	HBM2					187.39		93.69			NA	2	0	2

Source: UBS research, company data

Note: AMD MI300 and Intel Gaudi/Gaudi 2 TFLOP specs are based on our own estimates



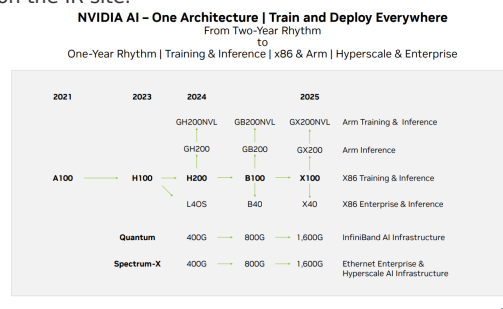
Source: UBS research, company data

Note: Intel Gaudi 2 and NVIDIA H100, H800 are restricted but outside the bounds of this chart (therefore not displayed)

3QF24 Update:

A lot again to like on the fundamentals here, mainly the reported data center demand strength across a wider aperture: half from CSPs (hyperscalers and GPU-specialized players) and the other half from consumer internet (e.g. DLRMs at Meta) and enterprises (e.g. software players such as Adobe, ServiceNow, Databricks, and Snowflake). A cyclical rebound was seen in gaming to 2x the pre-covid levels. NVDA also highlighted sovereign AI infrastructure spending across nations such as France, India, Japan, and Sweden.

- 3Q Datacenter revenues grew 41% q/q, reaching \$14.5B (80% of total revenues), also helping overall gross margins scale to 75% (non-GAAP). Networking assumed a \$10B annualized revenue run rate, with a growing set of ethernet solutions alongside Infiniband (+5x y/y) attached to the HGX. Recurring software/services are on track to exit this year at a \$1B annualized revenue run rate (e.g. via DGX Cloud and AI enterprise software).
- The inference thesis described in my original post seems to be playing out (with a convergence of training/inference onto H100s, with dedicated inference products such as L40S and GH200, and with Tensor RT LLM facilitating a 2x performance improvement and H200 another 2x over H100). The data processing pain points addressed here were discussed in-depth by Jensen on the call.
- China exposure was likely de-risked, with demand from the rest of the world more than offsetting the impact of recent restrictions. Approval of production/sales of H20 (China-focused regulation-compliant chip announced recently) constitutes an optionality.
- Beyond the top-line growth, you can't miss the substantial operating leverage in the model: ~64% operating margin and ~82% incremental operating margin (non-GAAP). Unadjusted FCF margins are ~34%, which we think should inch higher over the next few years in this fabless model.
- The likelihood of data center revenue growth in CY2025E is an open debate. It's too early to tell with certainty, but our research on AI adoption indicates that we are likely still early in a multi-year infrastructure spending cycle. This could likely be proven over time given a potentially catalyst-rich cycle across its products, including H200, Grace (CPU and superchip GH200), networking, and software/services. NVIDIA accelerated the cadence for new products from 2 years to 1 year, with the next generation GPU Blackwell B100 expected in CY2024E. Please see the graphic below from their October 2023 investor deck on the IR site.



Source: NVDA Investor Presentation, October 2023, <https://investor.nvidia.com/>

- One could argue it should be well-known at this point that they have tough comps coming up next year. The bear case of this ticker being the next pre-dot com bubble Cisco and the associated downside is probably overstated given the valuation ~25x on buy-side NTM earnings estimates (aggregated from desk analyst surveys). We continue to view that (i) hyperscaler efforts to develop custom silicon are not likely a major threat for external workloads, given the CUDA-accustomed 4M developer ecosystem moat and (ii) inference could grow from here into a larger market, with ample room for multiple GPU and accelerator providers. The risks to my thesis are plenty and are detailed in the original post. Please also see the disclaimer there.

Happy Thanksgiving!

Subject What a call
Entry 02/05/2024 10:03 AM
Member dynamicmoats
VIC needs more pitches like these. Differentiated insights about technology, well timed, insightful why now, large cap
big congrats

Subject Re: What a call
Entry 02/05/2024 01:13 PM
Member ahnuld
note the low rating it received and the growth investors club tag. Our members need to approach write ups with a more open mind me thinks

Subject C4Q23 Update: Inference
Entry 02/22/2024 07:23 AM
Member differentiatedfractal31415

A lot to like again here. Quick highlights:

C4Q23:

- Revenue: Datacenter \$18.4B Total \$22.1B
- Gross Margin
 - GAAP: 76%
 - Non-GAAP 76.7%
- Operating margin
 - Incremental margin: 77%
 - GAAP: 61.6%
 - Non-GAAP: 66.7%
- FCF margin :
 - 46% (unadjusted)
 - 51% (adding back SBC)

C1Q24 Guidance:

- Revenue: Total \$24B +/- 2%
- Gross Margin
 - GAAP: 76.3%
 - Non-GAAP 77%
- Operating Expenses
 - GAAP: \$3.5B
 - Non-GAAP: \$2.5B

The inference thesis detailed in my write-up is playing out faster than I had anticipated last March. I modeled inference reaching 50% of annual datacenter accelerated compute in 5 years; Collete mentioned that inference was 40% of revenues in CY2023 – for use cases such as deep learning recommendation systems (DLRS) and gen AI multi-modal inference from text to audio, image and video. As I discussed before, about 2/3 of inference workloads were estimated to be done on CPUs pre-LLM- so the inference uptake makes sense.

NVDA's earnings call offered more support to my rebuttals for the 5 bear theses:

- First, training demand is not likely to fall off the cliff after CY2023. I personally think there's lot more multimodal usecase potential to tap. Management cited strong demand across the board from the early data center infrastructure build cycle as hyperscalers and enterprises adopt accelerated compute and gen AI. H200 is being ramped up in production and demand trends remain high. The new product B100 is expected be supply-constrained.
- The bear view that the handoff from training to inference is likely to show up as an air pocket didn't happen – 40% of revenue (perhaps even more on workload volume) already comes from inference. The convergence of training and inference architectures to one platform and the co-designed (hardware/software) systems (not just chips) advantage of NVDA couldn't be more clear as we look back at 2023.
- Inference being competitive is likely but my thesis has been that NVDA should get its fair share, (if not dominant like in training), along with AMD, other accelerators, and custom silicon.
- The denominator is not just data center capex; Jensen delineated sovereign governments forming a growth vector for the long-term thesis.
- Hyperscaler custom silicon is likely going to focus on specific (e.g. DLRS) and not general-purpose workloads. Management announced on the earnings call a partnership with Amazon AWS for the DGX Cloud. AWS was not included in a key partnerships slide in the GTC 2023 presentation. AWS/NVDA partnership on GH200 was announced in AWS re:invent in Nov 2023. This is the largest and most mature cloud player that built and deployed its own custom silicon (e.g. via Annapurna labs acquisition) for several years now.

China exposure remains a risk, albeit at mid single digit percentage of revenues in the datacenter for C1Q24E (guided to be similar to C4Q23). Gross margins were elevated (from lower cost of components), and expected to be back in mid-70s in C2H24E. Software could have more legs from here – reached the \$1B annualized revenue run rate. Networking reached \$13B annualized revenue run rate.

If you follow the sector research on data centers and IT infrastructure, I think we still remain quite early in the data center infrastructure build cycle. My model CY25E EPS goes to \$34. Per bbg, the 5-year average NTM multiple on this name is ~40x. You can discount it to whatever level based on your view on the cycle and macro/rates, and growth beyond CY25E. Risk reward seems reasonable in my view based on the current mosaic picture of supply/demand presented in earnings calls so far and my view on the likely earnings trajectory. You can stress test this if you are a bear and believe this is the next Cisco of 1999-2000, which I personally think would be a stretch this early in the cycle.

All that said, the risks to my thesis are plenty and are detailed in the original post. Please also see the disclaimer there. My views can change anytime given the rapid pace of developments in the sector.

Will be at SFO MS TMT conference next month and at GTC 2024. Will be great to meet up over bull/bear debates if any of you would be around.

Subject
Entry
Member

Re: C4Q23 Update: Inference
02/22/2024 11:16 PM
pcm983

congrats on the call here.

maybe a dumb question but how do you square this:

Sales to one customer, Customer A, represented 13% of total revenue for fiscal year 2024, which was attributable to the Compute & Networking segment. One indirect customer which primarily purchases our products through system integrators and distributors, including through Customer A, is estimated to have represented approximately 19% of total revenue for fiscal year 2024, attributable to the Compute & Networking segment. Our estimated Compute & Networking demand is expected to remain concentrated. There were no customers with 10% or more of total revenue for fiscal years 2023 and 2022

Thanks

Subject	CSP ROI
Entry	02/26/2024 03:07 PM
Member	kgalt

Do we know the CSP ROI, even roughly?

For example, an H100 costs \$40k, the amort period is 6 years, the \$/GPU/hour is \$X, the electricity costs \$0.17 / kwh, the chip consumes 700W, what is the ROI for the CSP?

I've taken a stab at it but wondering if anyone has numbers.

Further, in the most recent call, towards the end, Jensen talks about "Nvidia AI Enterprise" and says that he charges \$4,500 per GPU per year. Not sure which GPU he's talking about but does this significantly undercut his CSP customers?

Subject	recent interview of Jensen
Entry	10/13/2024 11:24 AM
Member	differentiatedfractal31415

Jensen on this public podcast—nothing new compared to what's discussed in previous earnings calls or GTC/Computex, etc., but a good overall summary of his view of the inference debates.

<https://x.com/BG2Pod/status/1845431420728246688>

Not investment advice, please see cyclical, secular, and other risks highlighted in the original posting from Apr 2023.

Subject	Re: recent interview of Jensen
Entry	10/15/2024 08:57 AM
Member	bafana901

I used to program in visual basic. I was forced to covert these programs to Python not knowing the first thing in Python. Before I knew it I had much faster programs and in addition I was able to write programs beyond my wildest capabilities. This experience had a profound influence on me and as a result I have added an AI bucket to my portfolio dedicated to AI stocks.

Despite my positive experience with AI, I am the first to admit that I know nothing. This forces me to listen to industry leaders like Jensen and piggy back off their ideas. No original thought here.

What I found useful from the interview that was flagged included

1) NVDA invented a "new" computer

2) There is a \$1trillion dollars of old computers which need to be ripped out and replaced. Jensen thinks we are only \$150bil out of \$1trillion into this process. Long runway ahead.

3) New IT spend will be on "new tech", not "old tech" where innovation has slowed because Moore's law no longer applies. Who would waste their money on old tech run by cpu's.

The runway for "new tech" looks great, but is it discounted in the valuation of the market leaders already? Who will rip out the old stuff and replace it with the new and are these companies cheap enough to buy? I am very interested in other thoughts.

Subject	Re: Re: recent interview of Jensen
Entry	10/15/2024 10:05 AM
Member	differentiatedfractal31415

Last summer I coded an AI research assistant chatbot using Open AI code interpreter/open source SLMs, and a vector db from hugging face. I had thought it would take me 1-2 months for doing this project. I am an average/sporadic developer but Open AI made it just so easy I could get one functional research assistant running in less than 6 hours over a weekend. That got me to a hands-on feel and higher conviction we are early in this AI cycle as this was a real ROI-generating use case. I also thought that developer productivity should go through the roof. If the occasional developer can figure this out on a weekend, advanced developers should see even higher (nonlinear scaling) impact. Where to go from here? Dev Ittycheria addresses this in a great interview on Squawk box this am - talking around 4:00 timestamp about value accrual from chip layer -> model layer -> app layer - a great listen! In my opinion, inference scaling beyond chinchilla should add a growth leg to the chip layer and should keep XPU demand scaling linear with FLOPS - valuations are another debate altogether, for another day.

<https://www.youtube.com/watch?v=UXmY1cijhJI&t=2s>

Subject Re: Re: Re: recent interview of Jensen
Entry 10/15/2024 05:51 PM
Member bafana901

I think we may be in the same boat, experiencing the power of AI to complete projects that felt impossible generates a feeling of wonder that makes one very bullish on AI. Maybe a derivative of the Peter Lynch effect and that is why I added an AI bucket to my portfolio.

My next project is develop an AI agent. Can you share what tasks your agent is able to perform? I do a lot of quantitative work and I want to explore whether the agent can do some of this for me.

Subject H100s now available
Entry 12/10/2024 05:24 AM
Member ahnuld

So im trying to think through this so figured id post publicly for feedback.

My company has a GCP account, but we are small time users so normally are forced to take whatever chips are available. One of my engineers has been trying to finish off fine tuning a diffusion model but hasnt been able to access H100s except sporadically which has been a roadblock.

This week for the first time ever a GCP rep has been emailing me saying "Hey we have H100s available for you know! Very rare! Tell me your specs so we can reserve one for you."

Naturally this makes me think we are finally seeing supply catch up to demand (short NVDA), but I am wondering if maybe this is because Blackwells are finally shipping? But then again just because a few blackwells have shipped shouldnt mean a big jump in GPU capacity, its still a small incremental move, so my weighted average thought is still supply is catching up to demand. What is your interpretation.

ps. DF and Bafana, I spend all day, every day looking at early stage AI companies so if you want to see example cos of what you are describing just let me know.

Subject Re: Re: Re: recent interview of Jensen
Entry 12/13/2024 12:33 PM
Member hbomb5
 differentiatedfractal31415

Thank you for the updates. Definitely something to learn considering I have little to no background in the field. Please share your latest perspective on the industry and also the competitive landscape if possible. Enjoyed your link on the light-matter update.

Subject Are we back in the Year 2000?
Entry 01/11/2025 09:02 AM
Member zach721

Not picking on NVDA....specifically...but expensive large-cap tech...

Cisco is -35% since 2000.....300x increase in revenue 1990-2000...The largest market cap in market at one point (but top 3) ...universally loved with 55% revenue growth...then June 2000 to 2002 -90% and has recovered over the last 25 years to be only down -35%.

***Below is June 2000 NY Times article on Cisco
 NY Times***

"THE MARKETS: Market Place; Rapid Growth Makes Cisco A New Leader"

By Alex Berenson

• June 13, 2000

At any moment, one company stands in the spotlight of the middle ring in the stock market's never-ending circus. It may not be the biggest corporation in the world, or the most profitable, but somehow it both mirrors and leads the market's broader action.

For most of the 1990's, as personal computers moved from expensive toy to ubiquitous tool, Microsoft was that company. But with the Internet now more important than the PC, and Microsoft caught in a brutal antitrust fight, the software giant has lost its leadership to Cisco Systems, the San Jose, Calif., company that makes much of the equipment that powers the Internet.

The shift took place between March 1999 and March 2000, as Cisco's stock tripled, swelling the company's market capitalization by \$380 billion, to \$560 billion. To put that number in perspective, in just one year Cisco gained more market value than Merck, Coca-Cola and General Motors combined have built in their corporate histories.

Cisco "is as close to a bellwether as we have," said Giri Devulapally, vice president of T. Rowe Price, a Maryland mutual fund company that owns more than 28 million Cisco shares.

Even compared with its market value, which ranks third behind General Electric and Intel, Cisco's importance is outsized. The company is often the most heavily traded stock in the United States, with 50 million or more shares regularly changing hands in a day. In May, Cisco slumped more than 20 percent after a negative Barron's article, and the Nasdaq composite index slumped with it.

At first glance, Cisco may appear to be Exhibit A for investors worried that stocks remain too expensive despite their recent swoon. Even with its slide, Cisco trades at more than 125 times its earnings, five times the average for the Standard & Poor's 500-stock index. To justify that valuation, Cisco's earnings must grow far faster than those of most other companies.

Can they? Among Wall Street analysts and institutional investors, the consensus answer is a surprisingly firm yes. First, the bulls point to Cisco's history. In its July 1990 fiscal year, Cisco reported \$70 million in sales and \$14 million in profit. This year, analysts are expecting it to report \$19 billion in sales and almost \$4 billion in profits, a 300-fold increase.

Even more surprising, Cisco's growth has actually accelerated over the last two years. Many newly public companies are able to post a year or two of strong sales growth off a small base, but their growth almost always slows over time, thanks to what investment professionals call the law of large numbers.

As they grow, companies saturate their markets, become more complex and difficult to manage, and face larger and more entrenched competitors. And the mathematics of compounding requires a company to post ever-larger dollar increases in sales just to keep its growth rate flat.

But for now, Cisco is defying the law of large numbers. Except for one small blip, its year-over-year sales growth has increased for nine consecutive quarters, to an amazing 55 percent in its most recent quarter, ended April 30. Cisco's quarterly sales are now almost \$5 billion, and its profits before special charges topped \$1 billion for the first time last quarter.

"For a company that is as big as they are, the growth rate that they are achieving is pretty impressive," said Bill Keithler, manager of the Invesco Technology fund, which owns nearly \$200 million worth of Cisco shares. "We wish every company had Cisco's fundamentals."

And Cisco's growth is likely to continue at better than 30 percent a year for the next five years, analysts say. Cisco already dominates the market for the equipment that companies use to set up their internal networking systems and connect to the Internet. But the company is pushing heavily to get into the even bigger business of providing the equipment used by the largest telecommunications companies.

That "service provider" market, which will top \$100 billion this year and is growing at more than 10 percent annually, is a huge opportunity for Cisco, said Stephen Koffler, an analyst at First Union Securities. It is dominated by Lucent Technologies and Nortel Networks, but Cisco has quickly won a foothold, he said.

"Cisco competes against a very broad group of companies, and their market share, especially in the service-provider segment, is still very low," Mr. Koffler said. "Their ability to take market share is what gives the company the potential to grow."

Mr. Koffler thinks that Cisco's annual sales will triple over the next five years, to more than \$50 billion. "If that sounds impossible, remember that Lucent is almost on a \$40 billion run rate" of annual sales, he said.

Martin Pyykkonen, an analyst at CIBC World Markets, said Cisco had the potential to dominate Lucent and Nortel in the service-provider business the same way it defeated Cabletron Systems and 3Com Corporation in the Internet equipment market.

For Cisco to prosper, "all that really has to happen is for the Internet to continue to grow," Mr. Pyykkonen said. "I think they can grow into the valuation and convince people that there is still upside to come over the next year."

To be sure, no one is arguing Cisco is cheap, and some analysts remain wary of the company's valuation.

"It's sort of the devil on one shoulder and the angel on the other," said Paul Sagawa, a research analyst at Sanford C. Bernstein & Company, of the tension between Cisco's valuation and its growth. "Under the old way the market worked, you'd say, Cisco, great company, fully valued here, maybe even overvalued relative to its expectations."

Now Mr. Sagawa thinks Cisco investors might be in for a stretch of slow performance. If the market for technology stocks has actually cooled significantly, "we're back to where Cisco has to surprise, and there's probably going to be some multiple contraction over time," he said, implying that Cisco's stock might fall even if its earnings increase as its price-earnings ratio shrinks. And unlike some other analysts, Mr. Sagawa thinks Cisco may have a harder time dominating the service-provider business.

But even he rates Cisco as an "outperform," Bernstein's highest rating. "As far as having it as a core holding, it's hard to argue against it," Mr. Sagawa said, because Cisco's equipment has become essentially the default choice for corporations looking to buy Internet equipment. "You can turn the keys over to them, and they will take good care of you."



Subject DeepSeek - game changer?
Entry 01/25/2025 09:09 AM
Member Motherlode

Is this about to disrupt everything? power demand seems especially vulnerable.

DeepSeek articles started hitting yesterday from everywhere. Amazingly, the Economist hopped on this already too. The Chinese have built an opensource LLM which is HIGHLY efficient. It seems like this open-source model uses a Mixture-of-Experts ("MoE") approach, which directs the GPU to the relevant subject content. I am making this up, but I THINK this means that if you ask ChatGpt how gravity works... ChatGPT will scour every piece of info in its model -

including - for instance - a query within Shakespeare to see if there is relevant content on the science of gravity. By comparison, DeepSeek will skip Shakespeare and limit the computing to relevant content. This can cut power consumption by 90% / query. Perplexity is already including components of the model in their latest iterations. It sounds like DeepSeek has found a way to produce AI with significantly reduced computational/power load by cutting out irrelevant compute load. Hyperscalers can copy this approach and significantly improve efficiency.

This feels like a sell first, think later moment for an incredibly crowded trade. The interview with the Perplexity CEO in the middle of the Youtube video (link below) is very interesting. this tech appears to be a legitimate breakthrough.

<https://www.youtube.com/watch?v=WEBiebbeNCA&t=2414s>

<https://www.economist.com/briefing/2025/01/23/chinas-ai-industry-has-almost-caught-up-with-americas>

China's AI industry has almost caught up with America's

But Google, with all its resources, was not in fact the first firm to emulate OpenAI. Less than three months after o1 was launched, Alibaba, a Chinese e-commerce giant, released a new version of ...

www.economist.com



<https://www.nytimes.com/2025/01/23/technology/deepseek-china-ai-chips.html>

<https://www.reuters.com/technology/artificial-intelligence/tiktok-owner-bytedance-deepseek-lead-chinese-push-ai-reasoning-2025-01-22/>

what would this mean? hard to know exactly. this could mean that companies cut their capex or it might mean their high ROIC capital stretches far further and they invest the savings?

- Reduced power demand?
- ability to reinvest the savings into other productivity?
- reduced demand for GPUs?

See below on description from Chatgpt

Yes, DeepSeek's Mixture-of-Experts (MoE) architecture is designed to enhance computational efficiency by activating only the most relevant subsets of the model's parameters for a given task, leaving non-relevant areas inactive. This selective activation reduces computational costs considerably, allowing the model to perform well while being frugal with computation.

Medium

For instance, in DeepSeek-V2, the model comprises 236 billion total parameters, but only 21 billion are activated for each token during processing. This means that instead of engaging the entire network, only the pertinent "experts" within the model are utilized, leading to more efficient computation.

arxiv.org

Similarly, DeepSeek-V3 employs an MoE system where, out of its 671 billion parameters, only 37 billion are activated for any given task. This approach ensures that computational resources are focused solely on the relevant areas of inquiry, enhancing efficiency and reducing unnecessary processing.

daily.dev

By implementing this MoE architecture, DeepSeek effectively minimizes computational load by engaging only the necessary components of the model pertinent to the specific task, thereby optimizing performance and resource utilization.

Subject Re: DeepSeek - game changer?
Entry 01/25/2025 07:59 PM
Member eigenvalue

See the article I posted in the topics section earlier today, I discuss this in great depth towards the end of it:

https://youtubetranscriptoptimizer.com/blog/05_the_short_case_for_nvda

Subject Re: Re: DeepSeek - game changer?
Entry 01/26/2025 04:58 AM
Member grizzlybear

Anyone else finding it ironic that essentially on the 5 year anniversary of Covid, a shocking release of something cooked up in a China lab has the potential to trigger a massive deflationary episode across the US?

Subject Re: Re: Re: DeepSeek - game changer?
Entry 01/26/2025 08:42 AM
Member agentcooper2120

Thought this was a good post on that \$5.5m figure.

https://www.interconnects.ai/p/deepseek-v3-and-the-actual-cost-of/comments?utm_source=post&utm_medium=web&triedRedirect=true

***Edit:** The "cited numbers" here mostly refers to the general reaction to the release among the general public, as DeepSeek explicitly states that the costs are not the full picture*

Subject Re: Re: Re: DeepSeek - game changer?
Entry 01/26/2025 11:25 AM
Member pcm983

So true grizzly!

Subject DeepSeek - winners/losers?
Entry 01/27/2025 12:46 PM

Member

Motherlode

So here are my thoughts.

The cost to train and operate AI models will come down. Medium-term: this will drive substantially more investment as 1) new trainers enter the game 2) those that are experimenting already run way more trials. 3) The experimentation creates new applications and breakthroughs that drive more business and consumer applications. overtime, I suspect that once we have a adopted this breakthrough it will spawn more demand and it could be massive.

that said:

for now, do we have too many NVDA GPUs out there? have we ordered to much power?

- if I was about to spend a company changing amount of capital on a new training, I would hit pause and evaluate if I can do it for a fraction of the cost later this year. I watched a talk about how Bloombergpt was created and its learnings. It's staggering to see a massive company adopting conservative training methods to avoid costs. if these costs are suddenly reduced massively, I strongly suspect a wave of experimentation is kicked off once the west has figured how to integrate the tech. I am positive that bloomberg would have run let training runs go on longer and done more if it was 1/10th cost. what could they have found in so doing?

- if I was an hyperscaler, I would have to wonder if I was massively overpaying for something that might not be nearly as expensive in the coming years, i would certainly watch new power deals to make sure budgets aren't blown. that said, it might make sense to sign them now. if you believe in my medium-term view, then there is a risk that power is even shorter than we expected.

so how does this play out? do the hyperscalers see a big drop in 2H demand which roars back in 26? The drop in demand is driven by the customers recalibrating and initially starting with a lower total spend to make sure it works? if so, this could immediately hit NVDA. I can't see how WDC isn't a massive winner on this as it will mean more training, more biz/consumer applications which create data which much be stored, especially as these models move to the edge.

Seems like you need to think really hard about hyperscalers and NVDA into 2H25. i don't know what to do. Power could see a lag in new long-term deals, but I suspect this breakthrough ultimately means far more demand. I don't see how this isn't a massive POTENTIAL positive for WDC. a) nand is used more than hbm in lower power gpus 2) way more trials will be run that need to be stored 3) innovation which sends AI to the edge will create an enormous amount of data that needs to be stored.

Subject

Entry

Member

Hi, Motherlode --

Re: DeepSeek - winners/losers?

01/27/2025 01:00 PM

bowd57

> I suspect this breakthrough ultimately means far more demand.

Yeah, but -- 10x more?

> if I was an hyperscaler, I would have to wonder if I was massively overpaying for something that might not be nearly as expensive in the coming years,

This is maybe quibbling, but they're not (as I understand it) overpaying for compute, it's not like H100s slow down or anything.

I'm reminded of the old "power to cheap to meter" fusion fantasy. If Deepseek was 100x or 1000x instead of 10x, would that change anyone's thinking?

Yours,

Bowd

Subject

Entry

Member

The 2021 bubble peaked on Jan 27 with GME. The 2024-25 bubble peaked today (or 24th to be mathematically precise). History does rhyme.

It feels a lot of air needs to come over the next 3-4 months out the AI bubble (SOUN, AI, 3778 JT, etc but also NVDA and AMD), power/cables names (POWL, CRDO, AAOI, Prysmian, 5803 JT), GPU substrates (3110 JT, 4062 JT), quantum (RGTI, QUBT, QBTS, IONQ), flying taxies (ACHR, JOBY), nuclear (OKLO, SMR), random moonshot (ASTS, LUNR, etc.) - here are my losers. Winners from more demand in the long-term? Perhaps the same names that benefitted (over the years and not overnight) from the explosive growth of cloud computing, starting with TSM and yes, memory, and semiconductor software.

Subject

Entry

Member

Deepseek/Back To Scaling Laws

01/27/2025 01:08 PM

bowd57

Hi, all –

I confessed my ignorance about LLM/neural/whatever scaling here:

https://valueinvestorsclub.com/idea/TESLA_INC/3291208721/messages/230222

, scroll down a bit until you get to the chart.

I'm sure I don't understand what's going on, but get the impression that Deepseek is of interest because:

1: Provenance, it came totally from left field, and

2: Performance, which I take to be 10x? Is that right?

If #2 is right, and my understanding of that chart is right, then what's the big deal? An order of magnitude cheaper/more compute doesn't buy you a lot.

Certainly someone will be kind enough to set me straight.

Yours,

Bowd

Subject	Re: DeepSeek - winners/losers?
Entry	01/27/2025 01:19 PM
Member	utah1009

Small local hardware, edge, networks = winners. (I wonder if I serendipitously own any?)

Subject	Re: DeepSeek - winners/losers?
Entry	01/27/2025 01:26 PM
Member	rookie964

Motherlode,

For those who don't follow WDC and how they generate cash, can you elaborate your comments below in simple terms? Thank you.

"I can't see how WDC isn't a massive winner on this as it will mean more training, more biz/consumer applications which create data which much be stored, especially as these models move to the edge."

Subject	Re: Re: DeepSeek - winners/losers?
Entry	01/28/2025 08:47 AM
Member	Motherlode

if the cost to train LLMs drops by 10x, then the number of training efforts probably goes up by at least 5x... but Jevon's paradox... It probably ends up being 15x. This massively cuts the cost of entry and "democratizes" the tech to much smaller entities. For now, the only entities that can spend \$500mm+ to train a LLM are the mega-caps. if the cost becomes \$10-50mm, a) the existing AI giants will run far more trials b) more and more enterprises will experiment. All of this training creates data that needs to be stored. if we accelerate AI training, this SHOULD create more breakthroughs. These breakthroughs could have transformative applications to our lives. if so, these models will push out to the edge and the inference storage will explode.

In addition, if we use lower powered/cheaper GPUs to train, these GPUs use more nand than HBM. if you watch this youtube clip, you will see how carefully Bloomberg prepared to avoid training costs. as you are watching, think about all the different steps they might have taken if the training cost was 1/10th. what might they have found if they could have run 5-10x more trials? what might that have meant for demand if they had discovered something valuable?

[BloombergGPT: How We Built a 50 Billion Parameter](#)

Subject	Ben Thompson FAQ
Entry	01/28/2025 01:32 PM
Member	bowd57

Hi, all –

<https://stratechery.com/2025/deepseek-faq/>

A nice summary, albeit with a bit of auto-backpattigation for his brilliant insights. He also discusses stock market winners and losers!

A couple of snippets:

"we have a situation where AI models are teaching AI models, and where AI models are teaching themselves. "

The singularity is nigh!

Apropos of Google being a loser:

"any change from the status quo is probably a net negative."

Wrangler, this is what I was getting at "the terrifying prospect of de-moatification." Thompson thinks, and I think you agree, and I'm *not* disagreeing, that FB is a "winner" here. But things weren't so bad before! Big tech, FB included, has gotten so moaty and monopolistic that I think you have to start wondering if *any* change is for the better.

Yours,

Bowd

Subject	Re: Re: DeepSeek - winners/losers?
Entry	03/21/2025 09:52 AM
Member	mtmutombo

It feels a lot of air needs to come over the next 3-4 months out the AI bubble (SOUN, AI, 3778 JT, etc but also NVDA and AMD), power/cables names (POWL, CRDO, AAOI, Prysmian, 5803 JT), GPU substrates (3110 JT, 4062 JT), quantum (RGTI, QUBT, QBTS, IONQ), flying taxis (ACHR, JOBY), nuclear (OKLO, SMR), random moonshot (ASTS, LUNR, etc.) - here are my losers. Winners from more demand in the long-term? Perhaps the same names that benefitted (over the years and not overnight) from the explosive growth of cloud computing, starting with TSM and yes, memory, and semiconductor software.

Great comment, gtw! I am short most of these and think many of these have material downside. I'm a bit late to your comment since I don't read VIC much.

I will add the following for the group's consideration on the short side:

ASPI, BBAI, FFAI, KULR, LWLG, RXRX, VERI. All are AI-related (allegedly in most cases) or nuclear-related, which is an AI derivative. All of these are penny stocks prone to squeezes and may not be for everyone, but I believe these are all worthless or close to worthless except BBAI, so 100% downside for all of these is possible. The nice thing about these as shorts is aside from short-term timing, you don't actually have to be right about AI capex since you're short a basket of companies with no business models. Same with the quantum stocks gtw listed above. It's a bag of hot air.

Does VIC have any others I should be looking at? Thank you.

Subject	Re: Re: Re: DeepSeek - winners/losers?
Entry	03/21/2025 12:05 PM
Member	angus309

mtmutombo...thank you for the comments and list. Curious if you don't mind sharing why BBAI is not 100% downside in your opinion?

I don't have much to add other than other than NNE, another nuclear / SMR.

Subject	Re: Re: Re: DeepSeek - winners/losers?
Entry	03/21/2025 12:14 PM
Member	gtwt

Thanks mtmutombo! some other names are MOD, HPS/A CN, SEI and INOD. This speculative group lives its own life in terms of performance cycles - sometimes they underperform the markets, and then are resilient for a while, like over the past two weeks.

I'd separately highlight 5803 JT - it is subject to the same trends as POWL/MOD/HPS, but doesn't report orders or backlog and trades in a less efficient market, so a laggard relative to the group. Can trade funny at times though.

Love your penny stock group.

Subject	Re: Re: Re: Re: DeepSeek - winners/losers?
Entry	03/21/2025 12:27 PM
Member	mtmutombo

Great list, thank gtw. I will go through these and post any follow ups for new names not discussed in this thread if I find any. My impression is there are two buckets to really dig into. The first is the 'obvious' (thematic) AI names we're discussing here. The second bucket is names where there hasn't been a material AI impact to the business yet and the company is not ostensibly part of the 'AI bubble' (your mileage may vary on that term), but which are secular losers over the next 5+ years. In my experience, the market tends to underprice secular decline risk at big technology turning points, and I think there will be material downside in the right pockets with different factor exposure (value / value trap vs. meme / growth). Pockets might be consulting, call centers, etc. I haven't done work on the second bucket yet.

Subject	Re: Re: Re: Re: DeepSeek - winners/losers?
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**Entry
Member**

03/21/2025 12:35 PM
mtmutombo

Hi angus309, I should probably revise my statement and say BBAI *could* be a zero but it's actually a real business versus the other names I listed. The company has consistently negative operating cash flow, which is how stocks zero out in small caps in my experience (dreams are temporary, cash flow is forever). That said, it's a business with \$150m of revenue, which beats the pants off the other names I listed. They have legitimate government contracts (potentially a source of risk these days).

The principle issues with BBAI in my opinion are:

- 1) It's not *really* an AI stock. It has AI in the name and some low tech AI applications.
- 2) Despite this, it's perceived by retail investors as "the next Palantir" on social media. It is definitively not the next Palantir.
- 3) The PE sponsors are aggressively selling. There was a media campaign here to issue flakey PRs to run the stock up. Shares are down by 2/3rds off the recent peak, but insiders are still aggressively unloading around current levels.

Could it zero? Sure probably but it's not an obvious zero like LWLG, VERI, etc. that have obvious liquidity problems. I do like the setup where insiders compete for exit liquidity with a company's need to eventually raise capital though because the market can usually only absorb so many secondaries and when insiders sell en masse, it usually makes it harder to attract new institutional capital to the company.

After writing this out, maybe I was too generous to BBAI in my original comment!

Subject 1FQ2026 Update and AI 2025 from here: Scaling is (almost) all you need
Entry 05/29/2025 12:14 AM
Member differentiatedfractal31415

NVIDIA's 1FQ2026 results are in line with the scaling hypothesis I posted in the [COHR thread](#) in two comments: one on Dec 22, 2024 "Scaling is all you need" where I discussed reasoning inference with test-time compute (vs one-shot querying). and another on Jan 3, 2025 where I discussed model commoditization ('Dumb model paradox' thanks to DeepSeek and other model distillation trends) and value accrual to not just the app ('agent') layer but also the infrastructure layer (unlike the 'Dumb pipe' or 'Dumb cloud' paradoxes).

Amid fears of DeepSeek and other optimizations significantly impairing compute needs, the industry saw a step function increase in demand from reasoning. This was discussed by hyperscalers recently: Satya mentioned in 1Q25 earnings call 50T tokens processed by MSFT Azure in March (5x y/y in 1CQ25), Sundar cited at Google i/o that 450T tokens were processed last month.

- On the call, Colette mentioned hyperscalers deploying 1000 NVL72 racks/week on an average i.e. 72k GPUs per week, which is in the vicinity of 3.6M GPU top-4 customers' 2025 orders mentioned by Jensen in the GTC 2025 keynote
- NVDA had a Rule of 125+ print with total revenue 69%/y, FCF margin ~59%. Data center revenue grew 73%/y, of which Blackwell was 70%. This growth was achieved despite China headwinds from lost H20 revenues. The low gross margin 61% of the quarter was well-understood given the H20 dynamic, and came ahead of consensus. Networking rebounded, with NVIDIA's ethernet solution, Spectrum-X at \$8B annual run rate revenue
- The guidance for July quarter \$45B could have been \$53B (76%/y) if H20 China sales weren't banned. Views toward an improving Blackwell yield support achieving the mid 70s gross margin late C2025E

AI capex should eventually slow down at some point. Should there be a period of digestion? At some point, of course, especially if hyperscaler slow down may not be offset enough by the rising Sovereign spending. Should another DeepSeek-like innovation further accelerate unit inference cost declines? Yes, like it has done since 2022 (see LLMflation chart from A16Z). But to think we are done with the infrastructure build is probably a stretch. In line with the view from my original post in 2023 that inference would be a much bigger market, I think we have a long runway for scaling tokens (AI adoption across more industries, and with more AI usage per task) and at increasing token intensity (given the trend of increasing context windows and multi-step reasoning, offset by efficiency gains such as kv-cache optimization by NVDA Dynamo.)

Subject Re: 1FQ2026 Update and AI 2025 from here: Scaling is (almost) all you need
Entry 05/29/2025 04:07 AM
Member zamperini

Congrats. Nicely done.

Subject Re: 1FQ2026 Update and AI 2025 from here: Scaling is (almost) all you need
Entry 05/29/2025 10:43 AM
Member ahnuld

Hey DF,

On the 8 billion deferred china revenue, didnt this just get shifted around to other customers? I mean, if Blackwell chips are stocked out and were pre-ordered a year in advance, instead of shipping to china they prob shipped to other customers who preordered and are in the queue. If thats right, it would mean they didnt really "lose" 8 billion in sales in the coming quarter.

On overall demand. Do we really know deepseek impact on training demand yet? Again, if chip orders are places a year in advance and data center plans are made a year in advance, this wouldnt be showing up in the numbers yet. We'd prob start to see an impact in calendar 2026.

Would be glad to be told where I am wrong.

Subject	Re: Re: 1FQ2026 Update and AI 2025 from here: Scaling is (almost) all you need
Entry	05/29/2025 12:18 PM
Member	Wells

Chiming in here that the H20s were "purpose-built for the China market" (Jensen 03/05/25) and the chip "does not have a market outside of China" (Jensen last night)... to your point, would be easy to refill any displaced demand for B100s, but not the H20s. The \$4.6B charge for write-down on H20 inventory + purchase obligations was lower than the previously estimated \$5.5B, but because "we were able to reuse certain materials" (Colette last night) versus unloading some of the same inventory elsewhere.